Quasimaps and BPS counts

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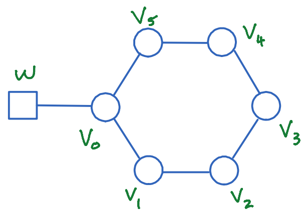
3d $\mathcal{N}=4$ quiver gauge theories. and quasimaps



Quiver Q

Higgs branch $\mathcal{M}^H_\theta(Q)$ = Nakajima quiver variety

(an algebraic symplectic reduction $T^* \mathrm{Rep}(Q) /\!\!/\!\!/_\theta \, GL(\vec{V})$)

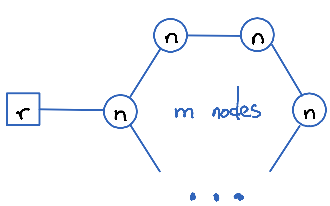
Carries an action of $GL(W) \times \mathbb{C}^\times_\hbar \supset A \times \mathbb{C}^\times_\hbar$

Coulomb branch $\mathcal{M}^C(Q)$ = BFN construction.

3d mirror symmetry (going back to [Intrilligator - Seiberg]) :

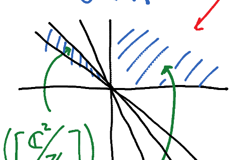
Quiver	Q	$\longleftrightarrow$	$Q^\vee$		
Higgs branch	$\mathcal{M}^H(Q)$	$\longleftrightarrow \mathcal{M}^C(Q^\vee)$	$\mathcal{M}^H(Q^\vee)$		
Coulomb branch	$\mathcal{M}^C(Q)$	$\longleftrightarrow \mathcal{M}^H(Q^\vee)$	$\mathcal{M}^C(Q^\vee)$		
FI parameters	$K \ni \vec{\theta} = (\theta_1, \dots, \theta_m)$	$\longleftrightarrow$ mass params of $Q^\vee$	$(\theta^\vee_1, \dots, \theta^\vee_m) = \vec{\theta}^\vee \in K^\vee$		"Kähler torus"
mass parameters	$A \ni (a_1, \dots, a_r)$	$\longleftrightarrow$ FI params of $Q^\vee$	$(a^\vee_1, \dots, a^\vee_r) \in A^\vee$		"Equivariant torus"

Our main example (affine type A quiver) :



$\vec{\theta} \in K$

"GIT chamber"

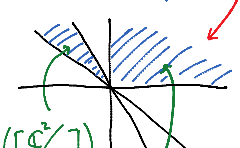


$\mathcal{M}^H_{\vec{\theta}} = \mathcal{M}_{r,n}([\mathbb{C}^2/\mathbb{Z}_m])$

$\mathcal{M}^H_{\vec{\theta}} = \mathcal{M}_{r,n}(\mathcal{A}_{m-1})$

$\vec{a} \in A$

"attracting chamber"



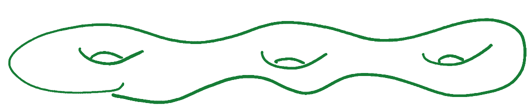
$\mathcal{M}^C_{\vec{a}} = \mathcal{M}_{m,n}([\mathbb{C}^2/\mathbb{Z}_r])$

$\mathcal{M}^C_{\vec{a}} = \mathcal{M}_{m,n}(\mathcal{A}_{r-1})$

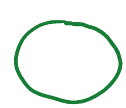
($\mathcal{M}_{r,n}(X)$ = moduli of rank r instantons on X of instanton number n.)

$r \leftrightsquigarrow m$ is like level-rank duality.

Can compute SUSY indices for these 3d theories.

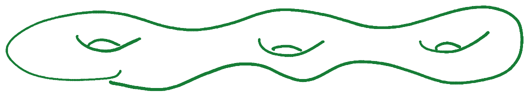


$\times$



space C time S^1

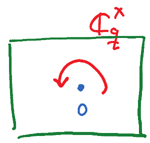
Do it in the IR limit, where $\mathrm{vol}(C) \gg 0$.
Contributions localize to (after a topological twist)

$$\left[\ \right.$$

$$\left. \text{space } C \xrightarrow{f} \mathcal{M}^H \right] \in \mathrm{QMaps}(C \to \mathcal{M}^H)$$

These are quasimaps (to $\mathcal{M}^H$).

(Twisted) SUSY index = Index of a virtual Dirac operator on $\mathcal{M}^H$ = $\chi\left(\mathrm{QMaps}(C \to \mathcal{M}^H), \hat{\mathcal{O}}^{\mathrm{vir}}\right)$

By degenerating C, get pieces

$$\left[\ \right.$$

$$\left. \mathbb{C} \subset \mathbb{P}^1 \xrightarrow{f} \mathcal{M}^H \right] \in \mathrm{QMaps}(\mathbb{P}^1 \to \mathcal{M}^H)_{\text{nonsing } \infty}.$$

(These depend on a boundary condition $\mathcal{E} \in K(\mathcal{M}^H)$ (at $\infty \in \mathbb{P}^1$))

$\Rightarrow$ indices are built from equivariant K-theoretic integrals on $\mathrm{QMaps}(\mathbb{P}^1 \to \mathcal{M}^H)_{\text{nonsing } \infty}$

using the symmetrized virtual structure sheaf $\hat{\mathcal{O}}^{\mathrm{vir}}$

(and possibly more insertions from $K(\mathcal{M}^H)$)

Quasimap vertex $V^{\mathcal{E}} = \chi\left(\underset{\text{with b.d. cond. } \mathcal{E}}{\mathrm{QMaps}(\mathbb{P}^1 \to \mathcal{M}^H)_{\text{nonsing } \infty}}, \hat{\mathcal{O}}^{\mathrm{vir}} \cdot z^{\deg}\right)$

Highly nontrivial power series in $\vec{z} = (z_1, \dots, z_m)$ ("Kähler variables")
valued in rational functions of $\vec{a} = (a_1, \dots, a_r)$, ("equivariant variables")
q and $\hbar$.

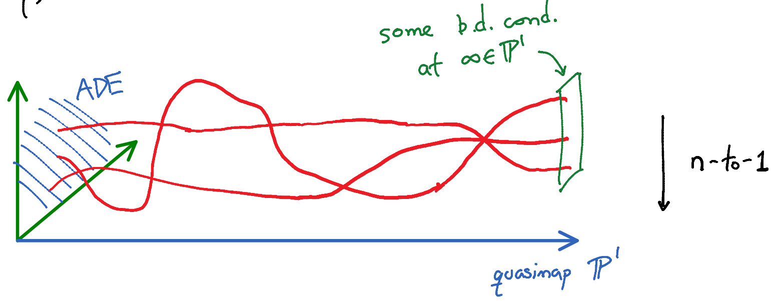
"CY limit"

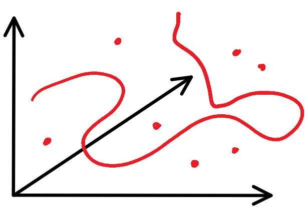
Useful: $q\hbar = 1$ $\Rightarrow$ all equivariant variables disappear.

Quasimaps and Donaldson–Thomas theory

Q = affine ADE quiver (1-dim framing, for simplicity) $\rightsquigarrow$ $\mathcal{M}^H$ = some flop of Hilb(ADE surface).

Graph Γ of a quasimap f
= 1-dim. subscheme
embedded in ADE $\times \mathbb{C}$!





Counting embedded subschemes in 3-folds
= (some flavor of) Donaldson – Thomas theory

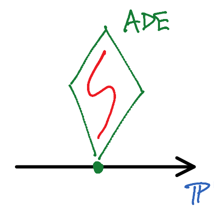
moduli Hilb(X, curves) of 1-dim subschemes in X.

not necessarily pure 1-dim.

BUT ! DT $\neq$ Quasimaps

(I.) has free-floating points

(II.) curves may have components lying completely in a fiber



Same principle as for quasimap vertex

$$V_{DT}(\text{ADE} \times \mathbb{C}) \neq V_{QMaps}(\text{ADE} \times \mathbb{C})$$

Removing unwanted contributions :

Option A: use this

$$\frac{V_{original}}{V_{unwanted}} = V'$$

Option B: construct theory immediately producing

Donaldson – Thomas [MNOP '03] $\rightsquigarrow$ remove (I.) Pandharipande – Thomas [PT '09] $\rightsquigarrow$ remove (II.) Bryan – Steinberg [BS '16]

V_{DT}

$V_{PT} \underset{\text{(conjectural)}}{=} V_{DT} / V_{DT,\,pts}$

$V_{BS} \underset{\text{(conjectural)}}{=} V_{PT} / V_{PT,\,fibers}$

for crepant resolutions only

MAIN THEOREM [H.L. '20] : (only type A in detail)

Whenever comparable (ADE surface fibrations over curves) :

Bryan–Steinberg theory = quasimap theory

isomorphic moduli spaces

\+ equivalent deformation theories (for producing $\hat{\mathcal{O}}^{vir}$)

at least in K-theory.

E.g.

$$V_{QMaps}(\mathrm{Hilb}(\mathcal{A}_{m-1})) = V_{BS}(\mathcal{A}_{m-1} \times \mathbb{C})$$

minor change of variables

$m=0$

$$\Rightarrow \quad V_{QMaps}(\mathrm{Hilb}(\mathbb{C}^2)) = V_{BS}(\mathbb{C}^2 \times \mathbb{C}) = V_{PT}(\mathbb{C}^3)$$

previously known (straightforward)

= (1-legged) topological vertex in CY limit

all equivariant variables disappear

$V_{QMaps}(X)$ $\updownarrow$ 3d mirror sym. $V_{QMaps}(X^\vee)$	$\xLeftrightarrow{QMaps/BS}$	$V_{BS}(Y)$ $\updownarrow$??? $V_{BS}(\text{"}Y^\vee\text{"})$

Thm [Aganagic–Okounkov] :

choice of attracting chamber

If $(X, \mathcal{C})$ is 3d mirror to $(X^\vee, \mathcal{C}^\vee)$,

Kähler ↔ equivariant

elliptic stable envelope.

appropriate normalization

$$\widetilde{V}_{QMaps}(X) = \widetilde{\mathrm{Stab}}^{Ell}_{\mathcal{C}^\vee} \cdot \widetilde{V}_{QMaps}(X^\vee)$$

(i.e. taking the CY limit on X

simplify by taking the CY limit on X

no equivariant variables on X

$$V^{CY}_{QMaps}(X) = S^{\bullet}\left(T^{1/2,\,C^\vee} X^\vee\right)\cdot 1$$

no Kähler variables on $X^\vee$

plethystic exp. : $S^{\bullet}(x) = \frac{1}{1-x}$

Example 1:

$$V^{CY}_{QMaps}\left(\mathrm{Hilb}(A_{m-1})\right) \overset{QMaps/BS}{=} V^{CY}_{BS}\left(A_{m-1}\times\mathbb{C}\right)$$

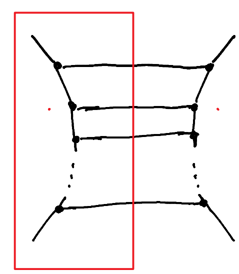
a bunch of topological vertices

3d mirror sym. ||

$$S^{\bullet}\left(T^{1/2,\,C^\vee}\mathcal{M}_m(\mathbb{C}^2)\right)$$

//

geometric engineering of pure $SU(m)$ 4d gauge theory [Iqbal - Kashani-Poor]



"half" of Nekrasov partition function.

Example 2:

$$V^{CY}_{QMaps}\left(\mathrm{Hilb}(A_{m-1})\right) = V^{CY}_{BS}\left(A_{m-1}\times\mathbb{C}\right) \quad \left(= \frac{V^{CY}_{PT}(A_{m-1}\times\mathbb{C})}{V^{CY}_{PT}(A_{m-1})} \right)$$

(as rational functions) //

??

$$V^{CY}_{QMaps}\left(\mathrm{Hilb}([\mathbb{C}^2/\mathbb{Z}_m])\right) = V^{CY}_{PT}\left([\mathbb{C}^2/\mathbb{Z}_m]\times\mathbb{C}\right)$$

DT crepant resolution conjecture [Bryan - Cadman - Young]

absorbed here

GIT chamber $C^\vee_-$

$$V^{CY}_{QMaps}\left(\mathrm{Hilb}(A_{m-1})\right) = S^{\bullet}\left(T^{1/2,\,C^\vee_-}\mathcal{M}_m(\mathbb{C}^2)\right)$$

equal up to $\underbrace{\text{overall monomial}}$

GIT chamber $C^\vee_+$

$$V^{CY}_{QMaps}\left(\mathrm{Hilb}([\mathbb{C}^2/\mathbb{Z}_m])\right) = S^{\bullet}\left(T^{1/2,\,C^\vee_+}\mathcal{M}_m(\mathbb{C}^2)\right)$$

$$T\mathcal{M}_m(\mathbb{C}^2) \overset{\text{in CY limit}}{=} \mathcal{G} + \mathcal{G}^\vee$$

(a very desirable property of K-theory classes called <u>self-duality</u>)

$\Rightarrow$ changing $C^\vee_+ \leftrightarrow C^\vee_-$ is like

$\Rightarrow$ changing $\mathcal{C}_+^\vee \leftrightarrow \mathcal{C}_-^\vee$ is like

$$S^\bullet(x) \leftrightarrow S^\bullet(x^{-1})$$

$$\frac{1}{1-x} = (-x)\frac{1}{1-x^{-1}} \qquad \frac{1}{1-x^{-1}}$$

(K-theory classes called self-duality)

Lift from CY limit:

Conjecture / Thm:
(K-theoretic DT CRC)

$$V_{PT}\left([\mathbb{C}^2/\mathbb{Z}_m]\times\mathbb{C}\right) = \tilde{R}^{Ell}_{\mathcal{C}_+^\vee \leftarrow \mathcal{C}_-^\vee} \cdot V_{BS}\left(A_{m-1}\times\mathbb{C}\right)$$

elliptic R-matrix $\quad \widetilde{Stab}^{Ell}_{\mathcal{C}_+^\vee} \cdot \left(\widetilde{Stab}^{Ell}_{\mathcal{C}_-^\vee}\right)^{-1}$

i.e. DT crepant resolutions governed by $\tilde{R}^{Ell}$!

Proof of MAIN THM

Via equivariant (virtual) localization,

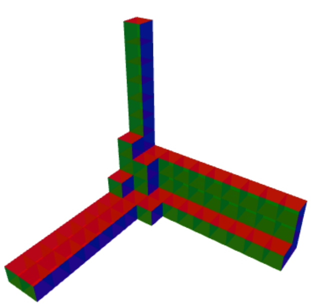
torus-fixed loci $\mathcal{M}^T$
+
virtual normal bundle $N^{vir}|_F$
at fixed components F

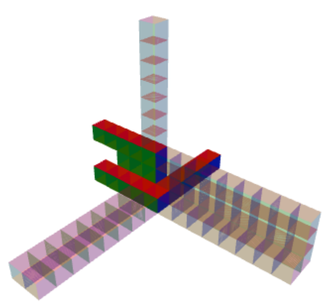
$\Big\} \rightarrow$ vertex V_μ (+ more)

$\Rightarrow$ 1. Match fixed points $BS_{\text{nonsing }\infty}(A_{m-1}\times\mathbb{C})^T \overset{\Phi}{\longleftrightarrow} QM_{\text{nonsing }\infty}(Hilb(A_{m-1}))^T$

2. Show that Φ preserves deformation theory.

Fixed points: DT PT BS

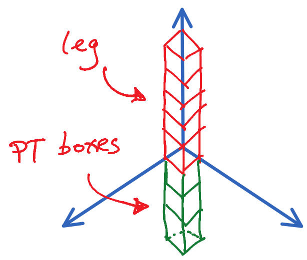




???

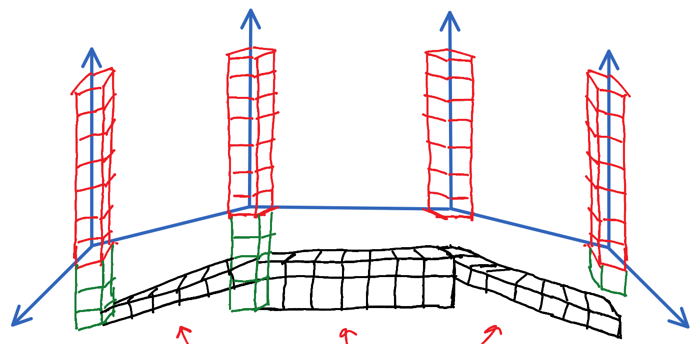
Note : Quasimap vertex $\leftrightsquigarrow$ DT/PT/BS 1-leg vertex

PT 1-leg vertex ($\mathbb{C}^3$)



$\rightsquigarrow$

BS 1-leg vertex ($A_3 \times \mathbb{C}$)

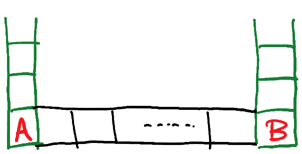


BS "rods"

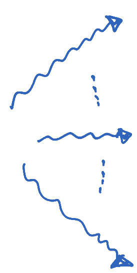
Each PT box can "shoot" left-/right-ward rods.

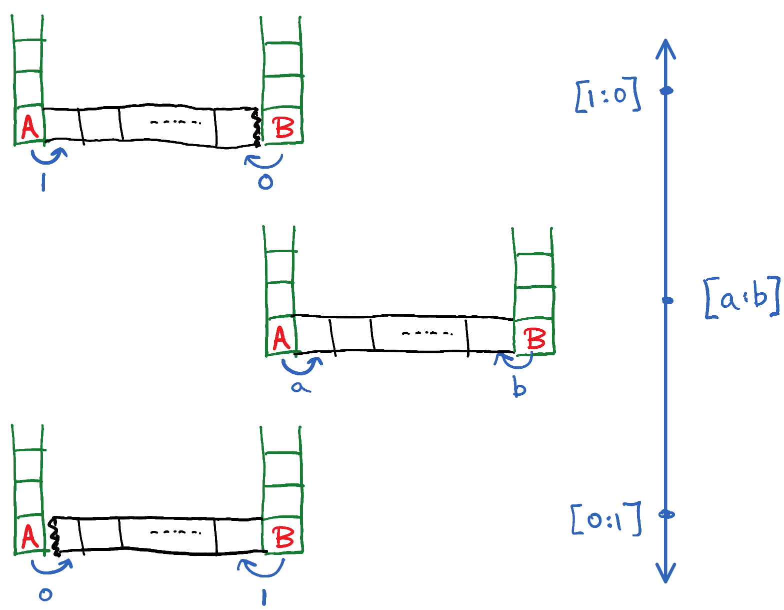
$\mathcal{O}_{E_i \cup \cdots \cup E_j}(0, 0, \ldots, 0, -1)$

There are positive-dimensional fixed loci !



Who shot the rod ?





Quasimap fixed loci are more standard (but still combinatorially involved)

$$BS_{\text{nonsing } \infty}(A_{m-1} \times \mathbb{C})^T \xleftrightarrow{\Phi} QM_{\text{nonsing } \infty}(\mathrm{Hilb}(A_{m-1}))^T$$

$$BS_{\text{nonsing } \infty}(A_{m-1} \times \mathbb{C})^T \overset{\Psi}{\longleftrightarrow} QM_{\text{nonsing } \infty}(\mathrm{Hilb}(A_{m-1}))^T$$

involved

straightforward combinatorially

(slightly less) involved

realized geometrically via derived McKay equivalence

$$D^b\mathrm{Coh}_{T/\Gamma}(A_{m-1}) \underset{\Phi}{\simeq} D^b\mathrm{Coh}_T(\mathbb{C}^2)$$

$\exists$ a way to encode fixed points on $\mathrm{Hilb}(ADE)$ as skyscrapers here.

Φ is an equivalence of categories

$$\Rightarrow \quad T^{vir}_{BS} \overset{\text{via } \Phi}{=} T^{vir}_{QMaps}$$

(thanks to A. Okounkov)

General stability arguments $\Rightarrow$ can move away from fixed loci:

$$BS_{\text{nonsing } \infty}(A_{m-1} \times \mathbb{C})^{\cancel{T}} \overset{\Phi}{\underset{\sim}{\longleftrightarrow}} QM_{\text{nonsing } \infty}(\mathrm{Hilb}(A_{m-1}))^{\cancel{T}}$$